Step 1: Add Title and Description

Add a Markdown cell in your notebook:

```
# Iris Flower Classification

This project uses the Iris dataset to train a simple machine learning model that can classify iris flowers into three species: Setosa, Versicolor, and Virginica.

We will use:
- pandas for data handling
- matplotlib and seaborn for visualization
- scikit-learn for building and evaluating the model
```

Step 2: Import Libraries

Add a code cell:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
```

Step 3: Load and Explore Dataset

```
iris = load_iris()
df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
df['species'] = iris.target
df.head()
```

Step 4: Visualize the Data

```
sns.pairplot(df, hue='species')

plt.show()
```

Step 5: Split the Data

```
X = df[iris.feature_names]

y = df['species']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Step 6: Train the Model

```
model = LogisticRegression(max_iter=200)

model.fit(X_train, y_train)
```

Step 7: Predict and Evaluate

```
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

print(classification_report(y_test, y_pred))
```

Step 8: Visualize Confusion Matrix

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=iris.target_names)

disp.plot(cmap='Blues')

plt.title("Confusion Matrix")

plt.show()
```

Final Summary (Markdown)

Project Summary: Iris Flower Classification

Built a simple ML model to classify iris flowers into Setosa, Versicolor, Virginica using Logistic Regression.

- Loaded and explored dataset

- Split into train/test sets
- Trained Logistic Regression model
- Evaluated with accuracy & confusion matrix
- Achieved 100% accuracy